SHARIF UNIVERSITY OF TECHNOLOGY · AI-NATIVE SYSTEMS DESIGN

Assignment 1 — A personal AI-native page

Individual · 1.5 points · Zahra Meskar, Taravat Niknam

What to build

Build a web page about yourself. A visitor opens it and asks a question. The page answers from a record you wrote, and it may also use an external API. The record is a JSON file with your courses, projects, and limits.

> The page may say only what is in your record, or what an external API returns. If a fact is missing from both, the page says it does not know, or it asks a question. It must not make up your biography.

Requirements

Part	What must be true
Intent	The visitor asks for a specific thing. Your page must handle at least three requests: summarize one project, suggest what to read based on your courses and projects, and say whether you fit a job the visitor describes. If the question is unclear, ask one question before you answer.
Context	Keep your courses, projects, and limits in a file the page reads. You may also call an external API and use what it returns. The model must not add facts from its own knowledge. Every fact in an answer must come from your file or from that API.
Schema and code quality	The JSON has a clear shape and a filled record. The code is clean and easy to read.
Capabilities	Give the model a short list of actions it may take: answer from your record or from an external API, refuse a fact you did not publish, and tell the visitor when the question is outside this page. It may not do anything else.
Authority	The page may talk about you. It may not send email, book a meeting, or speak for Sharif University of Technology. You may also set your own guardrails for the answers: topics you will not discuss, or facts you will not share. Write these limits on the page, and include them in the list of actions.
Other apps and APIs	Connect the page to at least one other app or API. Use what it returns when you answer. Say which service you used.
UI	Make the page clear and easy to use. A visitor can see where to type a question and where the answer appears.
Explainability	Show why the page answered that way. Name the fields it read and the action it took. If a fact is missing, say so.
Creativity	Add one idea of your own, beyond the items above.

Deploy on Hamravesh

The page must be online. A file that runs only on your laptop is not a submission.

1. Log in at console.hamravesh.com.
2. Read the docs at docs.hamravesh.com, including how Darkube creates and deploys an app. Follow those steps.
3. Create a new app in the console.
4. Deploy your page there. A visitor opens the public address and can ask the three questions.

Put the public address in your submission. Take the three traces from the live page.

What to submit

1. **The public address** of the app on Hamravesh, and the commands or steps you used.
2. **Your JSON file**: the schema (the shape of the data) and the filled record. Include courses, projects, and limits. Mark which fields a visitor is allowed to see.
3. **Three traces** from the live page, in a short report: a good answer, a refusal, and a clarifying question.

A trace is a short log of one question. Write the visitor's question, the intent the page recognized, the fields it read, the action it took, and the text it showed.

```
Question: Summarize my compiler project.
Intent: project_summary
Fields read: projects[compiler].summary, projects[compiler].stack
Capability: answer_from_record
Answer: The compiler project is a course compiler for a small language.
        The record lists a parser and a register allocator. It does not
        list a deployment date, so none is given.
```

This example uses a made-up project. Your three traces must come from your own page and your own record.

The three traces

1. **Good answer.** Ask something your record can answer. Name the fields the answer used.
2. **Refusal.** Ask for a fact you did not put in the record, such as a grade, a phone number, or an employer. The page refuses. It does not guess.
3. **Clarifying question.** Ask something vague, such as "Tell me about my project," when the record has more than one project. The page asks one follow-up. It does not pick a project for you.

Grading

Part	Points
Deploying the page: it is live on Hamravesh, and a visitor can open it	0.30
Schema and code quality: a clear JSON shape, a filled record, and clean code	0.30
Integration: the page uses other apps or APIs	0.30
Good UI: the page is clear and easy to use	0.20
Explainability: the page shows why it gave that answer	0.30
Creativity: you add an idea of your own	0.10
Total	**1.50**

Not accepted

- Your biography exists only in the system prompt. There is no separate record.
- An answer adds employers, grades, dates, or courses that are not in your record and did not come from an API you used.
- The page is not on Hamravesh, or it opens only on your computer.

We may hand you AI accounts. We will announce it later.

Due date: Tuesday, November 3, 2026.